SHOPEASE · BACKEND ENGINEERING

Product Service

Project Report

A FastAPI product catalog service with five CRUD endpoints, validated schemas, in-memory storage, and Docker packaging.

Python | FastAPI | Pydantic | In-memory | Docker

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Storage behavior: all product records are held in process memory. CRUD is available while the service runs; data clears when the process/container restarts. No database is integrated for this assignment.

01 · PROJECT OVERVIEW

Purpose and scope

ShopEase Product Service exposes a small, independent HTTP API for managing catalog products.

Clients send JSON requests to FastAPI routes. Pydantic validates request data, the route delegates to the product service, and a repository performs data access. The running application uses a thread-safe in-memory repository.

Core capabilities

- Create a product
- List, category-filter, and paginate
- Read one product by ID
- Replace product fields
- Delete a product

Operational goals

- Typed request and response schemas
- Stable REST status codes
- Containerized Uvicorn service
- Five complete CRUD operations
- Testable repository/service boundary

Product fields

Field	Purpose	Validation / storage
id	Product identifier	Database-generated integer
name, description	Display text	Up to 100 and 1000 characters
price	Unit price	Positive, at most 4 decimal places
stock_quantity	Available stock	Integer ≥ 0
category, status	Catalog grouping and state	Status: Active or Inactive
created_at	Creation timestamp	UTC timestamp generated by the service

02 · TECHNOLOGY AND STRUCTURE

Technology stack

Layer	Technology	Role
Runtime	Python 3.11+	Application language
HTTP API	FastAPI + Uvicorn	Routes, ASGI serving, OpenAPI docs
Validation	Pydantic	Request parsing and response shaping
Storage	Python dictionary + threading.Lock	Temporary process-local product collection
Container	Docker / Compose	Repeatable image and runtime
Verification	pytest, Swagger UI, Postman	Automated and manual API checks
Source control	Git / GitHub	Versioning and collaboration

Code organization

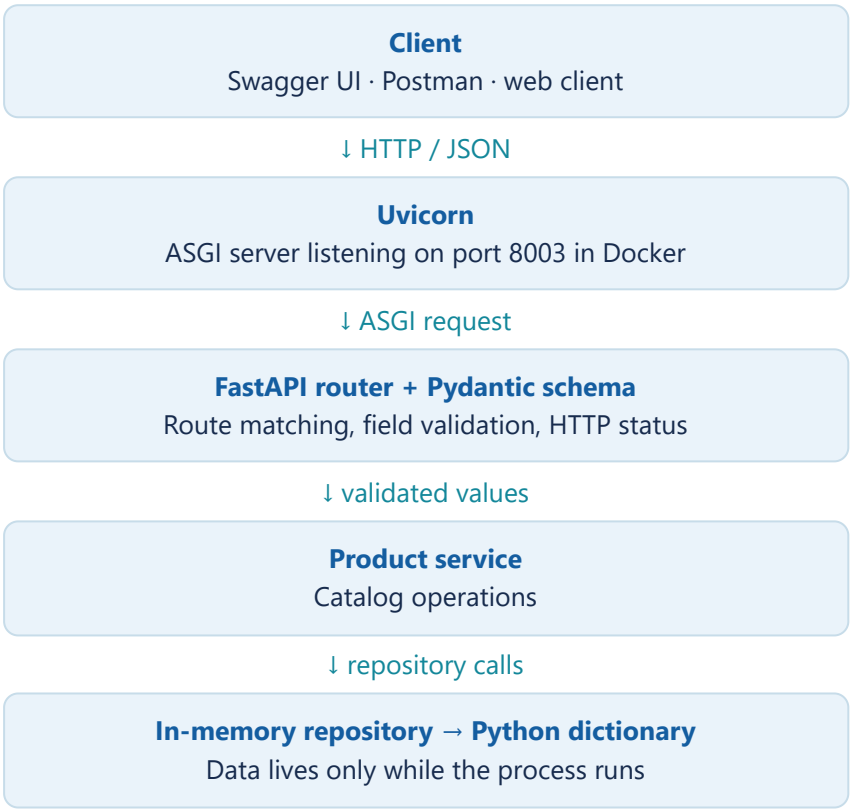
```
app/
  main.py                FastAPI app and health route
  config/settings.py     environment-backed settings
  routers/               HTTP endpoints
  schemas/               Pydantic validation
  models/                product data model
  services/              application-level operations
  repositories/          repository protocol + in-memory storage
  db/                    connection and migration runner
tests/                   API behavior and validation checks
Dockerfile / compose.yaml container build and run settings
```

> **Separation of concerns:** routes call the service, which delegates to the in-memory repository. This keeps HTTP behavior independently testable.

03 · ARCHITECTURE AND REQUEST FLOW

Layered request path

Each request passes through typed layers. Validation happens before business logic or a storage operation is attempted.



Response path

The repository returns a product model, the response schema shapes it, and FastAPI serializes JSON with the route's declared status code. Unexpected exceptions are logged server-side and return a generic 500 response.

Design benefits

- Input validation is consistent and visible in generated OpenAPI docs.
- The service layer provides a stable seam between routes and storage.
- The dictionary is protected by a lock for concurrent requests.
- Restart behavior is easy to understand: volatile product data is cleared.

04 · REST API DESIGN

Product endpoints

Method	Path	Behavior	Success
POST	/api/products/	Create product	201
GET	/api/products/	List / filter / paginate	200
GET	/api/products/{id}	Read by ID	200
PUT	/api/products/{id}	Replace product fields	200
DELETE	/api/products/{id}	Delete by ID	204

Create request example

```
{
  "name": "Headphones",
  "description": "Wireless headphones",
  "price": 25.50,
  "stock_quantity": 20,
  "category": "Accessories",
  "status": "Active"
}
```

List and error behavior

`GET /api/products/?category=Accessories&skip=0&limit=10` supports optional case-insensitive category matching and pagination. Unknown product IDs return 404. Invalid path or request data returns 422.

> Try endpoints interactively at `http://localhost:8003/docs` when the service is running. Use the same host and port in Postman.

05 · VALIDATION AND ERRORS

Guarding the API boundary

Pydantic schemas reject malformed or out-of-range input before it reaches the product service. Unexpected fields are forbidden so client mistakes are not silently ignored.

Input	Rule	Example rejected
name / category	Trimmed, non-empty, max 100 chars	Spaces only
description	Optional, max 1000 chars	Whitespace-only text
price	Strict positive number, max 4 decimals	0, negative, 0.00001
stock_quantity	Strict integer ≥ 0	-1 or a decimal
status	Active or Inactive	Pending
product_id	Integer greater than zero	0 or non-numeric text

HTTP status summary

200 OK
Read or update succeeded.

201 Created
Product created.

204 No Content
Delete succeeded.

404 Not Found
Product ID does not exist.

422 Unprocessable Entity
Invalid request, path, or query.

500 Internal Server Error
Unexpected server-side failure.

Storage boundary: all five product operations run against process memory. No database is required or integrated.

06 · IN-MEMORY STORAGE

Repository and data lifecycle

The application creates one in-memory repository at startup. It stores products in a Python dictionary indexed by generated integer IDs.

Operation	In-memory behavior
Create	Allocate the next ID, timestamp in UTC, add to dictionary
Read/list	Read dictionary; list supports category filtering and pagination
Update	Replace editable fields and preserve ID/creation timestamp
Delete	Remove ID from dictionary and report whether it existed

Concurrency and limits

A thread lock protects dictionary reads/writes. This storage is intentionally temporary: products disappear when the Python process or container restarts. No MySQL or other database is part of this assignment.

> **Important:** do not expect records to survive a container recreation. Use this implementation to demonstrate CRUD and Docker-hosted API behavior, not durable production persistence.

Price validation

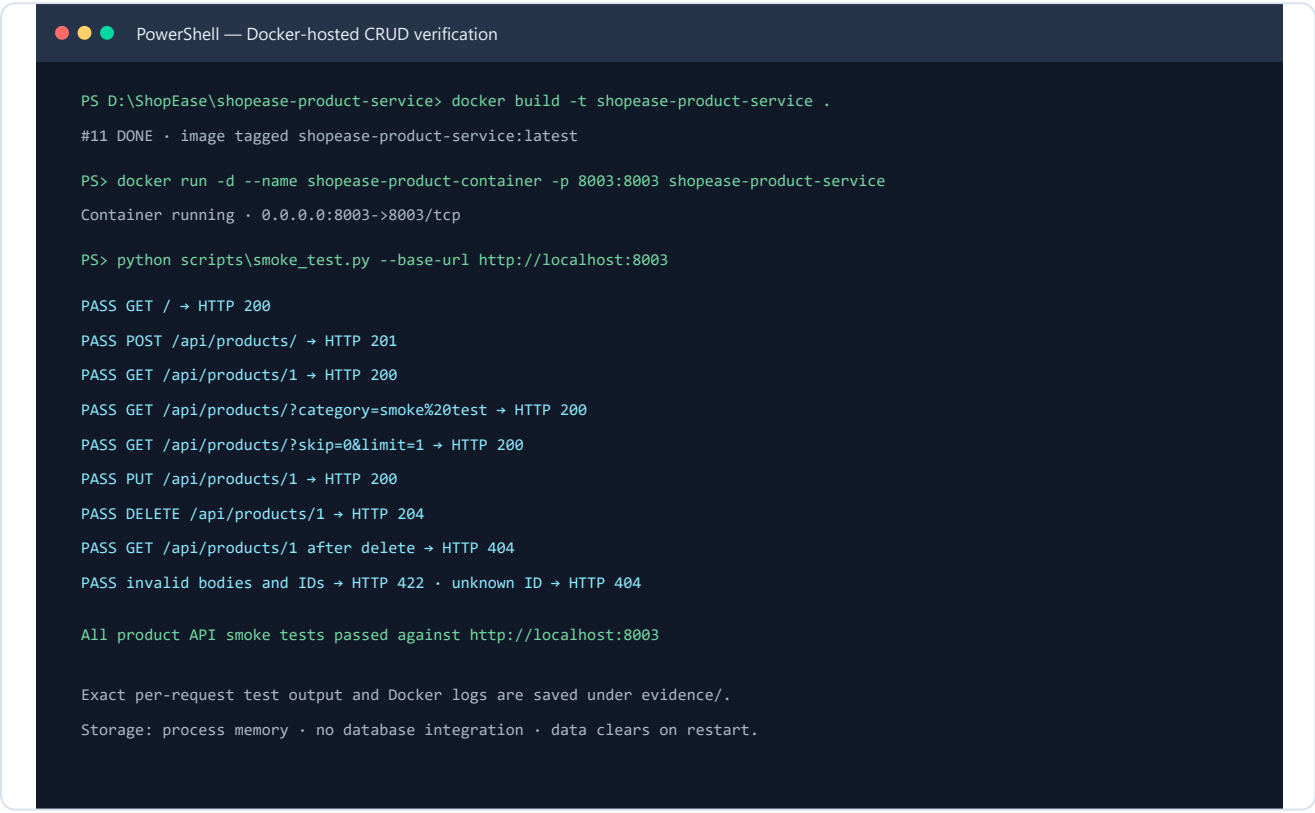
The API requires a positive price with at most four decimal places, and rejects invalid values before saving the product in memory.

07 · DOCKER BUILD AND RUN

Container output

The Docker image uses Python slim, installs the pinned application requirements, copies the app package, and starts Uvicorn on port 8003.

```
docker build -t shopease-product-service .
docker run -d --name shopease-product-container -p 8003:8003 shopease-product-service
docker ps
docker logs shopease-product-container
python scripts\smoke_test.py --base-url http://localhost:8003
```



Summary of the verified image, standalone container, and smoke test. Docker output and per-request results are saved under `evidence/`.

Where to test

- Health: `http://localhost:8003/`
- Swagger UI: `http://localhost:8003/docs`
- OpenAPI: `http://localhost:8003/openapi.json`

08 · LIVE API EVIDENCE

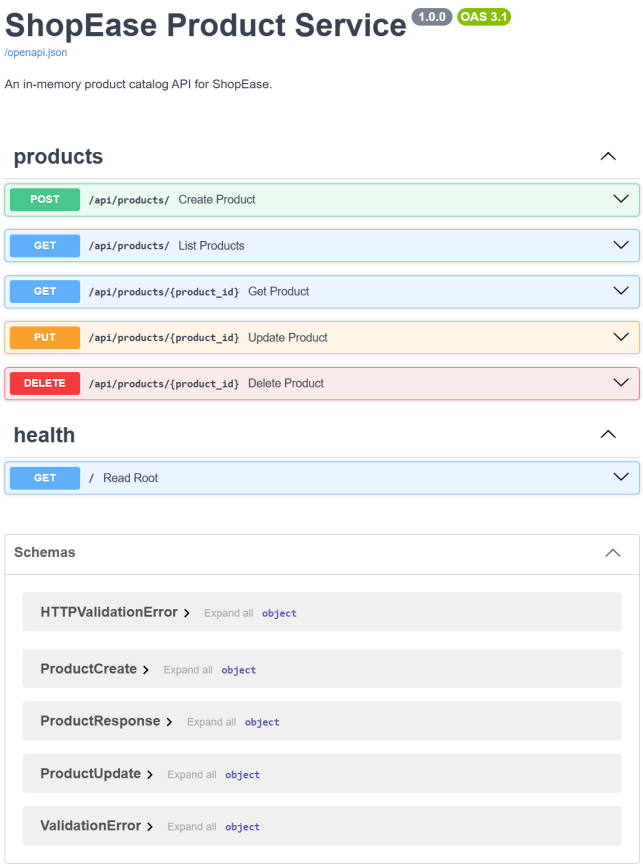
Running service responses

The standalone container was checked over the published local port. The screenshot below shows the actual Swagger UI exposed by the running container.

Pretty-print

{"message":"Welcome to the ShopEase Product Service"}

Actual browser screenshot from the live container; GET / returned 200.



Captured from http://localhost:8003/docs; it displays all five product operations.

09 · TESTING AND VERIFICATION

What passed and what remains

Check	Result	What it proves
pytest	18 passed	API CRUD behavior, validation, and in-memory default
pip check	Passed	No broken installed requirements
Python compileall	Passed	Application and test modules compile
Docker build	Passed	Image assembles with application dependencies
Container health	HTTP 200	App starts and root route responds
Swagger / OpenAPI	HTTP 200	Docs and schema endpoints are served
Memory lifecycle	201 → restart → 404	Confirms product data is intentionally temporary

Docker CRUD smoke test: passed against http://localhost:8003, including create (201), list/read/update (200), delete (204), invalid input (422), and missing/deleted IDs (404). Full per-request output is in `evidence/docker-crud-smoke-test.txt`.

Submission and team walkthrough

1. Review the source, this report, and the evidence files.
2. Each member should trace a request from route through schema/service to repository.
3. Demonstrate the five CRUD endpoints in Swagger or Postman.
4. Explain why data is cleared when the process restarts.
5. Commit and push the source and sanitized evidence to GitHub.

Team learning: every member should be able to trace a request from route to schema, service, and repository, then explain the returned HTTP status.